

# Surgical Instrument Trainer: Design and Evaluation of a Mixed Reality System for the Training of Operating Theatre Assistants

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Figure 1: Concept of mixed reality surgical instrument trainer

## ABSTRACT

Surgical education requires the combination of declarative knowledge with spatial, procedural, and sensorimotor skills, yet most training materials rely predominantly on two-dimensional media and passive instruction. This paper presents a mixed reality (MR) training system that enables learners to physically manipulate real surgical instruments while receiving spatially registered, context-aware digital guidance. The system is designed to support embodied learning and to improve the transition between training and clinical practice. In a controlled study ( $N = 42$ ), MR-based training was compared to conventional text- and image-based instruction. Results show that MR significantly improved recall of spatially embedded instrument knowledge, such as component names, while no overall advantage was observed for purely symbolic information. Both groups reported comparable gains in perceived learning success. User experience measures indicate high engagement, immersion, and minimal discomfort in the MR condition. These findings suggest that MR is particularly effective for learning tasks involving spatial relations and interaction, and may serve as a complementary modality to existing instructional approaches in early-stage surgical training.

**Index terms:** mixed reality, tacit knowledge, surgical training, embodied learning

## 1 INTRODUCTION

Most learning materials in medical education focus on the transmission of explicit, declarative knowledge and position learners primarily as recipients of theoretical content, such as in textbooks, lecture slides or video-based instructions.

In contrast, theories of situated learning and embodied cognition emphasize that medical expertise emerges through bodily embedded, context-dependent action [1] and is fundamentally shaped by material and social environments [2]. These essential components of medical competence including tacit knowledge in the sense of Polanyi [3], as well as procedural and sensorimotor skills, remain difficult to capture using existing learning materials and approaches. From a cognitive perspective, this limitation reflects a fundamental mismatch between learning materials and the conditions of knowledge application. According to the principle of Transfer-Appropriate Processing (TAP) [4], learning outcomes are optimized when the cognitive and sensorimotor processes engaged during training correspond to those required during later performance. Similarly, research on multimedia learning demonstrates that instructional formats which integrate interactive, multimodal, and spatially aligned information lead to superior learning outcomes compared to passive, text-based and image-based representations [5, 6].

This limitation is particularly pronounced in surgical education, where competence acquisition is inseparably linked to the physical manipulation of instruments, the rehearsal of specific working postures, and the coordination of fine-grained motor actions. Traditionally, such practice-based knowledge is conveyed through direct instruction and learning under real operating room conditions. While clinical learning remains essential, it is increasingly constrained by productivity demands, regulatory requirements, patient safety concerns, and ethical considerations. Physical simulations in skills laboratories on the other hand provide standardized and safe training environments but are limited in their ability to reproduce the situational, interprofessional, and affective complexity of real clinical practice. Mixed Reality (MR), which spatially aligns digital information with physical objects and supports embodied interaction, is particularly well suited to address this gap. It can align theoretical and practical learning more closely by combining declarative, procedural, and sensorimotor knowledge in learning scenarios. In particular, for early-stage training, these systems have the potential to support more practice-oriented and efficient

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competence development, reduce instructional burden, and contribute to improved patient safety.

## 2 RESEARCH QUESTION

MR training can bridge the gap between passive and interactive learning, two-dimensional and three-dimensional information, as well as explicit and tacit knowledge. By allowing learners to physically manipulate real surgical instruments while receiving context-aware digital guidance through visual overlays, MR can closely mirror real surgical workflows. We hypothesize that learners using MR will (H1) achieve higher gains in knowledge and (H2) report greater subjective learning success compared to learners engaging with traditional two-dimensional, text- and image-based materials. In addition, user experience, immersion, and any discomfort associated with MR use were assessed. Overall, we investigate whether MR-based interactive training improves knowledge acquisition, perceived learning success, and user experience compared to conventional 2D text- and image-based instruction.

## 3 SYSTEM DESIGN

The *Surgical Instrument Trainer* is a prototypical MR learning application designed to support the acquisition of instrument-specific procedural competencies in the training of operating theatre assistants (OTAs) (Fig. 1). OTAs play a critical role in surgical procedures, being responsible for the preparation, standardized organization, and correct handling of surgical instruments (Fig. 2). Consequently, instrument knowledge constitutes a core component of OTA training and must often be revisited during continuing training to accommodate procedure- and institution-specific workflows.

The *Surgical Instrument Trainer* is intentionally designed as an early-stage training system that targets basic instrument-related competencies prior to clinical participation. Its primary contribution lies not in conveying additional declarative information, but rather in coupling real instrument manipulation with spatially and temporally aligned digital guidance to support embodied, sensorimotor learning. By grounding instruction directly in physical interaction, the system should enable learners to practice actions related to handling, orientation, and handover that are difficult to acquire through text- or image-based materials alone. It is not intended to replace supervised clinical training or to support intraoperative use. Instead, the system aims to prepare learners for subsequent hands-on instruction by establishing a foundation of basic instrument knowledge, handling routines, and spatial familiarity in a controlled, low-risk setting.

The system design follows principles of embodied and situated learning by coupling physical interaction with spatially registered digital guidance. Real surgical instruments were used to preserve haptic fidelity, while MR overlays were employed to provide contextualized procedural information without abstracting material constraints. The application enables learners to explore the arrangement of surgical instruments in a three-dimensional and interactive manner using a video-passthrough virtual reality headset. Within the MR environment, trainees can see their real-world learning space and physical instruments, while digital guidance is overlaid to provide object-specific information and guide manual interaction with the instruments (Fig. 3). This approach allows learners to practice process- and context-related skills from an egocentric perspective and at an individual pace. From a didactic perspective, the *Surgical Instrument Trainer* follows a multi-stage learning concept: Building on the identification and selection of surgical instruments, learners can train the arrangement and the simulated handover of instruments to surgical staff. An integrated quiz mode supports targeted knowledge assessment and provides structured feedback for self-directed performance evaluation. In doing so, the demonstrator addresses both declarative and procedural knowledge components and supports autonomous navigation through practice-oriented training scenarios that facilitate embodied learning.



Figure 2: Instrument handling by an operating theatre assistant (OTA) during surgery.



Figure 3a-b: Mixed reality training scenario

## 4 METHOD

### 4.1 Software and Hardware

The implementation of the *Surgical Instrument Trainer* is available on GitHub [6]. The MR application was implemented in Unity (Unity Software Inc., San Francisco, USA) and deployed on a Varjo XR-3 headset (Varjo Technologies Oy, Finland), and integrated Ultraleap hand-tracking sensors (Leap Motion, Inc., San Francisco, USA). The choice of hardware was driven by technical constraints at the time of project initiation. Specifically, the Varjo XR-3 was the only commercially available MR system that combined high-quality video-passthrough with an open, low-latency integration pipeline for third-party optical hand-tracking. At the time of project initiation, consumer-grade standalone headsets offered passthrough and hand-tracking capabilities primarily optimized for general interaction, but did not yet provide the level of passthrough fidelity, calibration stability, and open integration with external optical hand-tracking systems required for interaction with physical surgical instruments.

## 4.2 Sample, Design, Procedure and Materials

Forty-two participants took part: 22 in the MR group and 20 in the paper-based control group, including 33 medical students and nine surgical assistant trainees. Sample size was determined via a priori power analysis ( $d = 0.8$ , one-sided  $t$ -test,  $\alpha = 0.05$ ,  $1 - \beta = 0.8$ ). A mixed  $2 \times 2$  design was used because a fully balanced within-subject design appeared unsuitable due to the limited number of available instruments that were trackable in the MR prototype. Learning mode (MR vs. 2D) was the between-subjects factor, and measurement time (before vs. after training) was the within-subjects factor. Participants first reported self-perceived knowledge of surgical instruments (H2), followed by two ten-minute training levels with subsequent tests.

In level one, participants learned names, components, and properties of ten instruments. Both groups interacted with the same set of real surgical instruments; the instructional difference lay exclusively in the presentation of guidance, which was delivered via spatially registered MR overlays in the MR condition and via printed 2D text-and-image materials in the control condition. The MR group used a Varjo XR-3 headset with interactive overlays, while the control group studied images and texts on printed sheets with identical information.

After a five-minute distractor task, participants' knowledge of the instruments (H1) was tested. They had to name the instrument and its components and indicate whether it can cause tissue trauma and has dangerous parts. Additionally, they had to locate the components' positions and dangerous parts on the real instrument. In level two, participants learned eight rules for surgical table placement. MR participants received visual, auditory, and feedback guidance via the headset; controls practiced independently using printed instructions. After another five-minute distractor task, participants' knowledge of the placement rules was tested (H1), both by asking them to state the rules and by measuring how correctly and accurately they were able to place the real instruments on a mock surgery table.

Then, to assess subjective learning success (H2), all participants answered the same questions on self-perceived knowledge as before the training. Additionally, the MR group filled out several scales that measured their experience with the MR training application.

## 4.3 Measures

Subjective learning success (H2) was operationalized as the mean difference in self-perceived instrument knowledge, which was measured before and after the training via five items, answered on a 5-point Likert-scale (e.g., "I am able to name the components of common surgical tools"). Objective instrument knowledge (H1) was operationalized as performance-based assessments at level 1 and 2. Multiple outcome measures were computed and compared between the MR and control groups: (a) the number of correctly named instruments; (b) the number of correctly named components; (c) the number of correctly located components; (d) the number of instruments correctly classified as traumatic or non-traumatic; (e) the number of correctly identified dangerous components; (f) the number of correctly recalled placement rules; and (g) the number of instruments correctly placed on the table. The MR group also completed the UEQ-Short [8] for pragmatic and hedonic quality, a five-item immersion scale (adapted from [9,10]), and a six-item cybersickness scale (adapted from [11]).

## 4.4 Analysis

Data were analyzed in R (version 4.4.2); significance was set at  $p < 0.05$ . H1 (objective knowledge) was tested using Welch  $t$ - or Wilcoxon tests to compare MR and control groups; H2 (subjective learning success) was assessed by comparing pre- to post-training changes. The  $p$ -values for H1 were adjusted using Bonferroni-Holm correction to account for multiple testing. Evaluation metrics for immersion, cybersickness, and training effectiveness were summarized using descriptive statistics. For evaluation of specific training aspects, descriptive values of single items were exploratively compared between the groups, but no significance was calculated. For the MR group, descriptive values of user experience, immersion and cybersickness were calculated.

## 5 RESULTS

With respect to learning outcomes (H1), participants trained with the MR system demonstrated significantly superior performance in recalling spatially and functionally embedded information (Fig. 4): The MR group recalled significantly more names of instrument components ( $M = 7.73$ ,  $SD = 2.23$ ) than the control group ( $M = 5.55$ ,  $SD = 2.67$ ),  $t(37) = 2.86$ ,  $p = 0.035$ ,  $d = 0.89$ . This advantage aligns with the MR system's spatially registered overlays and direct physical interaction with real instruments. There was a numerical tendency for the spatially unembedded instrument names to be recalled better in the control group ( $M = 5.95$ ,  $SD = 1.88$  vs.  $M = 4.82$ ,  $SD = 1.62$ ), but this effect was not significant,  $t(38) = 2.08$ ,  $p = 0.176$ ,  $d = 0.65$ . No further significant differences in learning outcomes were observed across both learning levels (all  $p > 0.05$ ).

Both groups reported substantial gains in self-perceived knowledge (H2). In the MR group, ratings increased from  $M = 1.85$  ( $SD = 0.71$ ) to  $M = 3.96$  ( $SD = 0.47$ ),  $t(21) = 13.62$ ,  $p < 0.001$ ,  $d = 2.90$ ; in the control group from  $M = 1.86$  ( $SD = 0.84$ ) to  $M = 4.10$  ( $SD = 0.48$ ),  $t(19) = 14.64$ ,  $p < 0.001$ ,  $d = 3.27$ . The pre-post change did not differ between groups,  $t(40) = 0.62$ ,  $p = 0.54$ , indicating that both instructional formats were effective in increasing subjective confidence. Exploratory evaluations highlighted MR-specific experiential differences. Participants in the MR condition tended to rate the training as more practical ( $M = 4.55$ ,  $SD = 0.74$  vs.  $M = 3.90$ ,  $SD = 1.21$ ) and expressed greater interest in using comparable MR systems for future surgical skills training ( $M = 4.50$ ,  $SD = 0.67$  vs.  $M = 3.80$ ,  $SD = 1.06$ ). In contrast, control participants reported slightly higher perceived competence after training ( $M = 4.30$ ,  $SD = 0.80$  vs.  $M = 3.91$ ,  $SD = 0.81$ ), potentially reflecting the familiarity and lower cognitive load of 2D-based learning.

Focusing on the MR condition, UEQ-S scores indicated very high hedonic quality ( $M = 6.41$ ,  $SD = 0.71$ ) and high pragmatic quality ( $M = 5.83$ ,  $SD = 0.61$ , both on a 7-point scale), suggesting that the system was perceived as engaging, innovative, and functionally useful. Item-level analysis showed the highest ratings for "interesting" ( $M = 6.50$ ,  $SD = 0.80$ ) and the lowest ratings for "easy" ( $M = 5.55$ ,  $SD = 0.80$ ), indicating a learning curve associated with MR interaction. Participants reported high immersion ( $M = 3.97$ ,  $SD = 0.61$  on a 5-point scale) and minimal discomfort ( $M = 1.52$ ,  $SD = 0.61$  on a 7-point scale), with visual strain as the most frequent but still low-intensity symptom ( $M = 2.05$ ,  $SD = 1.25$ ).

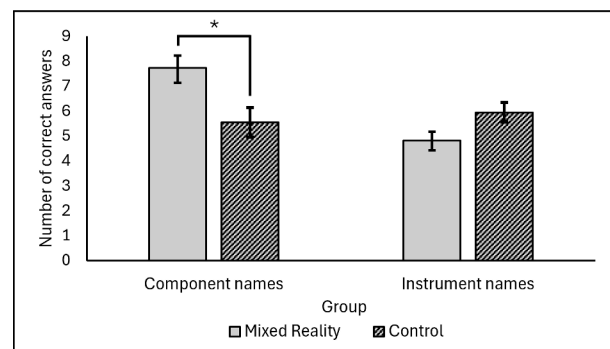


Figure 4: Number of correct answers in the first learning level. (\* indicates  $p < 0.05$ )

## 6 DISCUSSION

This study investigated whether MR-based training can support the acquisition of surgical instrument knowledge more effectively than conventional learning material based on 2D media. The results indicate that MR can afford certain benefits for spatially grounded learning, rather than a general advantage across all knowledge dimensions. Participants trained with MR demonstrated superior recall of instrument components,

supporting theories of embodied cognition and transfer-appropriate processing. By enabling physical manipulation of real instruments combined with spatially registered digital overlays, the MR system likely facilitated integration of perceptual and motor information.

Subjective learning gains were substantial and comparable across conditions. While MR did not increase perceived learning success relative to 2D-based learning material, this finding is notable given the novelty and prototype status of the MR system. Importantly, MR participants rated the training as more practical than conventional 2D-based training media and expressed greater willingness to use similar systems in the future, indicating high perceived relevance and acceptance. User experience measures further support feasibility: The system was rated as highly engaging and immersive, with minimal discomfort, though usability ratings suggest room for improvement in interaction design.

Overall, the findings indicate that MR can achieve learning outcomes comparable to established training methods, despite being a relatively novel instructional modality. While improvements were modest and primarily observed in tasks requiring spatial understanding, the results suggest that MR may be particularly well suited for supporting learning processes that benefit from three-dimensional representation and interaction. Rather than replacing existing approaches, MR appears to function effectively as a complementary modality and may offer advantages over conventional 2D media in contexts where spatial relationships and object manipulation are central to the learning objectives.

## 7 LIMITATIONS AND FUTURE WORK

This study has several limitations: Learning outcomes were assessed immediately after training, limiting conclusions about long-term retention and transfer. Moreover, while MR-based training benefitted naming instrument components, no difference in locating them was observed. This is surprising, as locating seems like an even more spatially-aligned task than naming. However, this finding could reflect a ceiling effect: Twelve components were named and localized, and both groups were able to correctly localize the positions of all components ( $M = 10.55$ ,  $SD = 1.85$  in the MR;  $M = 10.05$ ,  $SD = 1.88$  in the control group). Localization, which did not require correct naming, was therefore probably too easy to find a difference. Future studies should therefore use more complex instruments with more components to assess advantages of MR-based learning on localization tasks.

In addition, the MR system is a prototype that could only track a limited number of instruments and components. Therefore, the evaluation could only be carried out with a limited number of instruments and tasks. Certain hand poses while handling the instruments could cause occlusion, preventing the headset cameras from properly detecting the tracker. In some cases this could result in temporary tracking loss and the digital instrument overlay disappearing. During our study, we observed that users quickly adapted by adjusting their grip on the instruments. However, this could lead to unnatural hand poses, which may negatively impact perceived ergonomics and overall comfort.

Future studies using more mature MR systems with refined interaction design, a broader range of instruments, and increased task complexity may reveal stronger and more differentiated learning effects, particularly for spatially and procedurally demanding aspects of surgical training. More advanced training scenarios could address multi-user configurations involving expert guidance, procedural tasks, or time-critical scenarios that simulate operative stress. However, these features were deliberately excluded from the present study to isolate the effects of embodied, spatially registered guidance on basic instrument learning. While introducing social interaction, procedural complexity, or performance pressure could provide additional features, it would exceed the scope of the current evaluation.

We note that the present study does not aim to evaluate cost-effectiveness of specific hardware platforms. Instead, the XR-3 was employed as an enabling research instrument to isolate interaction design and training effects under optimal sensing conditions. Future work should explicitly investigate the

transferability of the proposed training approach to contemporary standalone headsets with lower acquisition costs.

## 8 DESIGN IMPLICATIONS

Based on the empirical findings, we derive the following design implications for MR-based surgical training systems: (1) MR can support certain learning strategies through spatial and structural interaction. Designers should prioritize spatially registered, interaction-driven content for knowledge acquisition, while carefully integrating symbolic information such as labels to avoid attentional neglect. (2) Overlays for non-spatial information (e.g., instrument names) should be coupled with user interaction. Dynamic presentation triggered by manipulation events or task progression may better support learning outcomes. (3) High immersion and hedonic quality can enhance motivation but may also increase exploratory behavior that distracts from learning goals. MR interfaces should guide attention toward task-relevant information, particularly in early learning stages to balance engagement and cognitive load. (4) While discomfort was minimal, lower ease-of-use ratings indicate the need for streamlined interaction techniques and onboarding to support scalable use in educational contexts.

## 9 CONCLUSION

This work contributes empirical evidence on the benefits of MR for spatial and embodied learning in surgical education. By allowing learners to physically manipulate real instruments while receiving context-aware digital guidance, the MR system improved recall of spatially embedded instrument knowledge, particularly instrument components, and supported embodied, interactive learning. Participants reported high engagement, immersion, and minimal discomfort, indicating that MR can improve early-stage surgical training for OTAs. While MR did not increase recall of purely symbolic information, these findings suggest that MR is most effective for tasks requiring spatial, haptic, and procedural understanding. Future work should explore long-term retention, transfer to clinical performance, and more complex or collaborative surgical scenarios to further validate MR as a complementary training modality.

## FIGURE CREDITS

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## SUPPLEMENTARY MATERIAL

A supplementary video demonstrating the Surgical Instrument Trainer and its interaction workflow is available online at: <https://youtu.be/rAYDi4JsbnM>.

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